



GRACE COLLEGE OF ENGINEERING
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MULLAKKADU, THOOTHUKUDI - 628 005

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SCIENCE

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Unit- II Problem Solving

Prepared By,

Mrs.R.Janani, AP/AI&DS

Unit- II Problem Solving

PART A

1. When is the class of problem said to be intractable.

The problems whose algorithm take an unreasonably large amount of resources (time and/or space) are called intractable.

For example -TSP

2. What is the power of heuristic search ? Why does one go for heuristic search ?

Heuristic search uses problem specific knowledge while searching in state space This helps to improve average search performance.

They use evaluation function which denote relative desirability (goodness) of expanding node set. This makes the search more efficient and faster. One should go for heuristic search because it has power to solve large, hard problems in affordable times.

3. What is the advantage of heuristic function ?

Heuristic function ranks alternative paths in various search algorithms, at each branching step, based on the available information, so that a better path is chosen.

The main advantage of heuristic function is that it guides for which state to explore now, while searching. It makes use of problem specific knowledge like constraints to check the goodness of a state to be explored. This drastically reduces the required searching time.

4. State the reason when hill climbing often gets stuck.

Local maxima is the state where hill climbing algorithm is sure to get stuck. Local maximum is the peak that is higher than each of its neighbour states, but lower than the global maximum. So we have missed the better state here. All the search procedure turns out to be wasted here. It is like a dead end.

5. When a heuristic function his said to be admissible ? Give an admissible

Admissible heuristic function is that function which never over estimates the cost to reach the goal state. It means that $h(n)$ gives true cost to reach the goal state 'n' The admissible heuristic for TSP is Minimum spanning tree.

i) Minimum assignment problem.

6. What do you mean by local maxima with respect to search technique ?

Local maximum is the peak that is higher than each of its neighbour states but lower than the global maximum ie, a local maxima is a tiny hill on the surface whose peak is not as high as the main peak (which is a optimal solution). Hill climbing to find optimum solution when it encounters local maxima.

Any small move, strong fails also makes things worse (temporarily). At local maxima all the search procedure turns out to be wasted here. It is like a dead end.

7. How can we avoid ridge and plateau in hill climbing ?

Ridge and plateau in hill climbing can be avoided using methods like backtracking, making big jumps. Backtracking and making big jumps helps to avoid plateau, whereas, application of multiple rules help to avoid the problem of ridges.

8. What is meant by Turing test?

To conduct this test we need two people and one machine. One person will be an interrogator (i.e.) questioner, will be asking questions to one person and one machine. Three of them will be in a separate room. Interrogator knows them just as A and B. so it has to identify which is the person and machine. The goal of the machine is to make Interrogator believe that it is the person's answer. If machine succeeds by fooling Interrogator, the machine acts like a human. Programming a computer to pass Turing test is very difficult.

9. Define Rational Agent. It is one that acts, so as to achieve the best outcome (or) when there is uncertainty, the best expected outcome.

10. Define Agent. An Agent is anything that can be viewed as perceiving (i.e.) understanding its environment through sensors and acting upon that environment through actuators.

11. Define an Omniscient agent. An omniscient agent knows the actual outcome of its action and can act accordingly; but omniscience is impossible in reality.

12. Give the structure of agent in an environment?

Agent interacts with environment through sensors and actuators. An Agent is anything that can be viewed as perceiving (i.e.) understanding its environment through sensors and acting upon that environment through actuators.

13. What is autonomy? A rational agent should be autonomous. It should learn what it can do to compensate for partial (or) in correct prior knowledge.

14. What is important for task environment?

PEAS → P- Performance measure E - Environment A- Actuators S – Sensors Example Interactive English tutor performance measure maximize student's score on test.

15. List the steps involved in simple problem solving technique.

i. Goal formulation ii. Problem formulation iii. Search iv. Solution v. Execution phase

16. What are the different types of problem?

Single state problem, multiple state problems, Contingency problem, Exploration problem.

17. What are the components of a problem?

There are four components. They are i. initial state ii. Successor function iii. Goal test iv. Path cost v. Operator vi. state space vii. Path

18. Define State Space.

The set of all possible states reachable from the initial state by any sequence of action is called state space.

19. Define Path.

A path in the state space is a sequence of state connected by sequence of actions.

20. Define Path Cost.

A function that assigns a numeric cost to each path, which is the sum of the cost of the each action along the path.

21. Give example problems for Artificial Intelligence.

i. Toy problems ii. Real world problems

22. Give example for real world end toy problems.

Real world problem examples: i. Airline travel problem. ii. Touring problem. iii. Traveling salesman problem. iv. VLSI Layout problem v. Robot navigation vi. Automatic Assembly vii. Internet searching

Toy problem Examples: Vacuum world problem. 8 – Queen problem 8 – Puzzle problem

23. Define search tree.

The tree which is constructed for the search process over the state space is called search tree.

24. Define search node.

The root of the search tree that is the initial state of the problem is called search node.

25. Define fringe.

The collection of nodes that have been generated but not yet expanded, this collection is called fringe or frontier.

26. Compare and Contrast Admissible and Consistent Heuristics.

- **Admissible Heuristic:** A heuristic is admissible if it never overestimates the true cost to reach the goal, ensuring that the solution found is optimal. It guarantees the correctness of the A* algorithm but doesn't necessarily improve efficiency.

- **Consistent Heuristic:** A heuristic is consistent (or monotonic) if, for every node and its successor, the estimated cost to reach the goal is less than or equal to the cost to reach the successor plus the

estimated cost from the successor to the goal. Consistent heuristics are always admissible and also ensure that the search is efficient by preventing redundant node expansions.

27.Comment on the statement “Breadth -first search is a special case of uniform-cost search”

The statement "**Breadth-first search is a special case of uniform-cost search**" is **true**.

- **Breadth-first search (BFS)** expands nodes level by level, assuming all edges have the same cost (typically cost = 1). It explores nodes in increasing order of their depth from the start node.
- **Uniform-cost search (UCS)**, on the other hand, expands nodes based on the **lowest total path cost** from the start node, ensuring the least-cost path is explored first.

When all edge costs are equal, UCS behaves like BFS because both algorithms prioritize nodes in the same order (by their depth). Therefore, **BFS is a special case of UCS when all edge costs are uniform**.

28. Define admissible and dominant heuristic.

Admissible Heuristic: A heuristic is admissible if it never overestimates the cost to reach the goal. In other words, it always provides a lower bound or equal to the actual cost of reaching the goal from the current state. This ensures the optimality of the solution when used in search algorithms like A*.

Dominant Heuristic: A heuristic is dominant if it is always at least as good as another heuristic in terms of the estimates it provides, and it is strictly better in some cases. A dominant heuristic leads to faster search as it guides the search more efficiently toward the goal.

29. What is the purpose of a contingency plan?

The purpose of a **contingency plan** is to outline the actions to be taken in response to unexpected events or emergencies. It ensures that an organization can quickly adapt, minimize disruptions, and maintain operations in the face of unforeseen circumstances or risks.

29. what are the things that agent knows in online search problems.

- The **current state**: The agent is aware of its present location or situation in the problem space.
- Action **outcomes**: The agent knows the effects of actions it takes but typically does not have full knowledge of the entire problem space. It makes decisions based on partial or limited information, often relying on exploration and feedback from the environment.

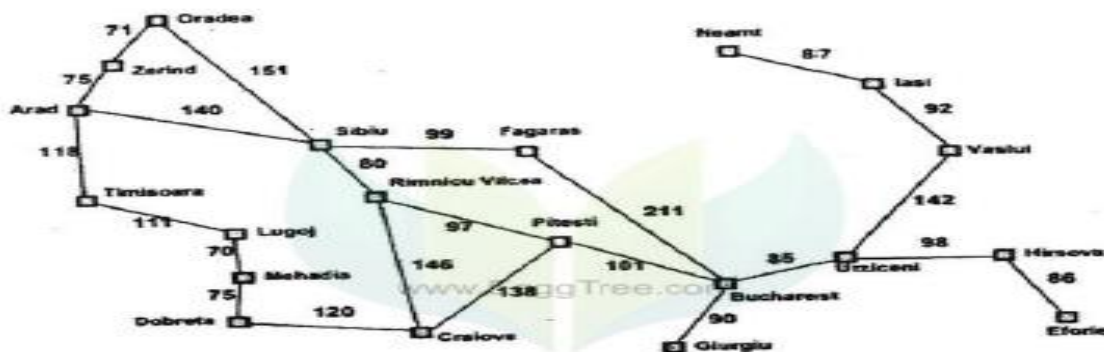
30. What is heuristic function.

A **heuristic function** is a function used in search algorithms to estimate the cost or distance from the current state to the goal state. It provides guidance to the algorithm by evaluating the desirability of different paths, helping it prioritize which actions or states to explore next in order to find the optimal solution more efficiently.

PART-B

- 1.Explain the heuristic search with example.
- 2.Explain in detail about the heuristic function.
- 3.What is an path cost? Explain the A* search with example.
4. Explain the properties Hill climbing tree search.
- 4.Explain the components of local search optimization problems with example.
- 5.Discuss informed search methods with example.
- 6.What is partially observable environments with examples.
- 7.Briefly explain the search strategies in online search agents.
- 8.What is unknown environments and explain with an examples.
- 9.

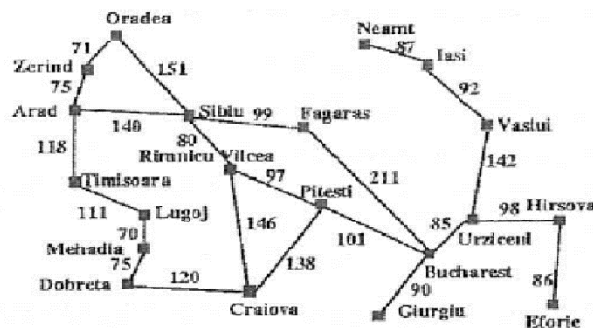
(b) Trace A* Algorithm to find the shortest route from Lugoj to Bucharest using the straight line distance heuristic for the below diagram and the straight line distance given in the table. Also elaborate on A* algorithm.



Town	Air Dist.	Town	Air Dist.
Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Dobreta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Lasi	226	Vaslui	199
Lugoj	244	Zerind	374

10.

Outline A* Algorithm. Trace the algorithm to find the shortest route from Lugoj to Bucharest using the straight line distance heuristic. Show the sequence of nodes traversed by the algorithm. The straight line distances are given below.



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11. How does online agent searches in Unknown environment.
12. What is heuristic search technique in AI? How does heuristic search works? Explain its advantages and disadvantages.
13. Describe the local search algorithm with neat sketch.
14. Explain the steps involved in formulating problems with example.
15. Write short notes on genetic algorithm.
16. Elaborate on the need for local search algorithm and discuss any one in detail.
17. Discuss on Online Search Agents that uses depth-first exploration.

